Ans. (d) : According to the section, the international community is yet to identify the migration category of migrants. Climate change has also led to migration within the country, which is being identified. Therefore, the most logical inference is that the relationship between climate charge and migration is not yet well understood.

Passage-7

Many farmers use synthetic pesticides to kill infesting insects. The consumption of pesticides in some of the developed countries is touching 3000 grams/hectare. Unfortunately, there are reports that these compounds possess inherent toxicities that endanger the health of the farm operators, consumers and the environment. Synthetic pesticides are generally persistent in environment. Entering in food chain they destroy the microbial diversity and cause ecological imbalance. Their indiscriminate use has resulted in development of resistance among insects to insecticides, upsetting of balance in nature and resurgence of treated populations. Natural pest control using the botanical pesticides is safer to the user and the environment because they break down into harmless compounds within hours or days in the presence of sunlight. Plants with pesticidal properties have been in nature for millions of years without any ill or adverse effects on the ecosystem. They are easily decomposed by many microbes common in most soil. They help in the maintenance of biological diversity, of predators and the reduction of environmental contamination and human health hazards. Botanical pesticides formulated from plants are biodegradable and their use in crop protection is a practical sustainable alternative.

22. On the basis of the above passage, the following assumptions have been made:

1. Synthetic pesticides should never be used in modem agriculture.
2. One of the aims of sustainable agriculture is to ensure minimal ecological imbalance.
3. Botanical pesticides are more effective as compared to synthetic pesticides.

Which of the assumptions given above is/are correct?

(a) 1 and 2 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Ans. (b) : One of the most important assumptions made on the basis of the above passage is that one of the objectives of sustainable agriculture is to ensure minimum ecological imbalance. Hence, option (b) is the only correct answer.

23. Which of the following statements is/are correct regarding biopesticides?

1. They are not hazardous to human health.
2. They are persistent in environment.
3. They are essential to maintain the biodiversity of any ecosystem.

Select the correct answer using the code given below.

(a) 1 only
(b) 1 and 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Ans. (a) : Of the given statements about biopesticides, only statement (1) is correct that they are not dangerous to human health.

Directions for the following 7 (seven) items:

Read the following seven passages and answer the items that follow the passages. Your answers to these items should be based on the Passages only.

Passage-1

An air quality index (AQI) is a way to combine measurements of multiple air pollutants into a single number or rating. This index is ideally kept constantly updated and available in different places. The AQI is most useful when lots of pollution data are being gathered and when pollution levels are normally, but not always, low. In such cases, if pollution levels spike for a few days, the public can quickly take preventive action (like staying indoors) in response to an air quality warning. Unfortunately, that is not urban India. Pollution levels in many large Indian cities are so high that they remain well above any health or regulatory standard for large part of the year. If our index stays in the Red/Dangerous' region day after day, there is not much any one can do, other than getting used to ignoring it.

24. Which among the following is the most logical and rational inference that can be made from the above passage?

(a) Our governments are not responsible enough to keep our cities pollution free.
(b) There is absolutely no need for air quality indices in our country.
(c) Air quality index is not helpful to the residents of many of our large cities.
(d) In every city, public awareness about pollution problems should increase.

Ans. (c) : According to the passage, "The situation in Urban India is not like this. Pollution level in many large Indian cities are so high that they remain above health standards as regulatory standards for most days of the year." The most air quality indices are not helpful for many residents of our big cities.

Passage-2

Productive jobs are vital for growth and a good lob is the best form of inclusion. More than half of our population depends on agriculture, but the experience of other countries suggests that the number of people dependent on agriculture will have to shrink if per capita incomes in agriculture are to go up substantially. While industry is creating jobs, too many such jobs are low-productivity non-contractual jobs in the unorganized sector, offering low incomes, little protection, and no benefits. Service jobs are relatively of high productivity, but employment growth in services has been slow in recent years.

25. Which among the following is the most logical and rational inference that can be made from the above passage?

(a) We must create conditions for the faster growth of highly productive service jobs to ensure employment growth and inclusion.
(b) We must shift the farm workers to the highly productive manufacturing and service sectors to ensure the economic growth and inclusion.
(c) We must create conditions for the faster growth of productive jobs outside of agriculture even while improving the productivity of agriculture.
(d) We must emphasize the cultivation of high-yielding hybrid varieties and genetically modified crops to increase the per capita income in agriculture.

Ans. (c) : Option (c) is the best answer that can be inferred logically this is what we should do now as a nation to absorb agriculture workforce and improve agriculture itself.

Passage-3

A Landscape-scale approach to land use can encourage greater biodiversity outside cdected areas. During hurricane 'Mitch' in 1998, farms using eco agricultural practices suffered 58 percent, 70 percent and 99 percent less damage in Honduras, Nicaragua and Guatemala, respectively, than farms using conventional techniques. In Costa, vegetative windbreaks and fencerows boosted farmers' income from pasture and coffee while also increasing bird diversity. Bee pollination is more effective when agricultural fields are closer to natural or seminatural habitat, a finding that matters because 87 percent of the world's 107 leading crops depend on animal pollinators. In Costa Rica, Nicaragua and Colombia silvopastoral systems t tint integrate trees with pastureland are improving the sustainability of cattle production, and diversifying and increasing farmers' income.

26. Which among the following is the most logical and rational inference that can be made from the above passage?

(a) Agricultural practices that enhance biodiversity can often increase farm output and reduce the vulnerability to disasters.
(b) All the countries of the world should be encouraged to replace ecoagriculture with conventional agriculture.
(c) Ecoagriculture should be permitted in protected areas without destroying the biodiversity there.
(d) The yield of food crops will be very high if ecoagricultural practices are adopted to cultivate them.

Ans. (a) : The most logical and rational inference that can be drawn on the basis of the section of Biodiversity is as follows - Agricultural practices that increases biodiversity often increases farm production and reduce the vulnerability to disasters.

Passage-4

The medium term challenge for Indian manufacturing is to move from lower to higher tech sectors, from lower to higher value-added sectors, and from lower to higher productivity sectors. Medium tech industries are primarily capital intensive and resource processing; and high tech industries are mainly capital and technology intensive. In order to push the share of manufacturing in overall GDP to the projected 25 per cent, Indian manufacturing needs to capture the global market in sectors showing a rising trend in demand. These sectors are largely high technology and capital intensive.

27. Which among the following is the most logical and rational inference that can be made from the above passage?

(a) India's GDP displays high value-added and high productivity levels in medium tech and resource processing industries.
(b) Promotion of capital and technology intensive manufacturing is not possible in India.
(c) India should push up the public investments and encourage the private investments in research and development, technology upgradation and skill development.
(d) India has already gained a great share in global markets in sectors showing a rising trend in demand.

Ans. (c) : The given passage is on 'Manufacturing in India'. We need to find the most logical and rational inference- that the idea of Indian manufacturing needs to go from low to high in the areas of productivity, technology and skill development.

Passage-5

Over the last decade, Indian agriculture has become more robust with record production of food grains and oilseeds. Increased procurement, consequently, has added huge of food grains in the granaries. India is one of the world's top producers of rice, wheat, milk, fruits and vegetables. India is still home the quarter of all undernourished people in the world. On an average, almost half of the total expenditure of nearly half of the households is on food.

28. Which among the following is the most logical corollary to the above passage?

(a) Increasing the efficiency of farm to-fork value chain is necessary to reduce the poverty and malnutrition.
(b) Increasing the agricultural productivity will automatically eliminate the poverty and malnutrition in India.
(c) India's agricultural productivity is already great and it is not necessary to increase it further.
(d) Allocation of more funds for social welfare and poverty alleviation programmes will ultimately eliminate the poverty and malnutrition in India.

Ans. (a) : India is an agricultural country. Most of the people here are dependent on agriculture. But today the farmer of the country is compelled to commit suicide. He is suffering from poverty and malnutrition. Therefore, it is necessary to increase the efficiency of the value chain from farm to plate to prevent poverty, malnutrition and suicide.